Introduction on UML for Industrial Systems

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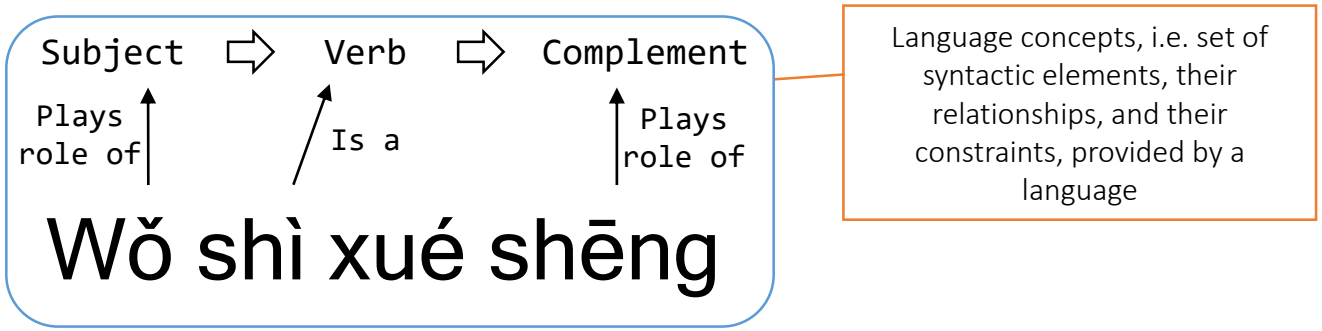
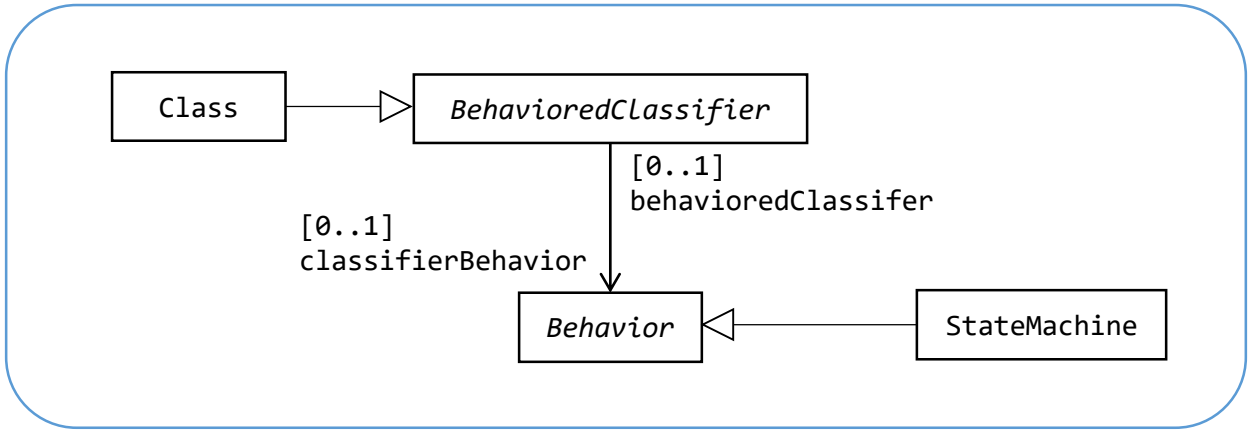
Domain specific modeling languages (DSLs)

1. **Recap – abstract syntax, notation, and semantics**
2. **What are domain specific languages, what are their objectives?**
3. **Text-based DSLs**
4. **UML profile mechanisms**
5. **Define a simple profile**

Abstract Syntax
The concepts defined by a language





Notation
The representation of language concepts

Symbolic representations associated to language concepts (i.e. elements of the abstract syntax)

我 是 学 生

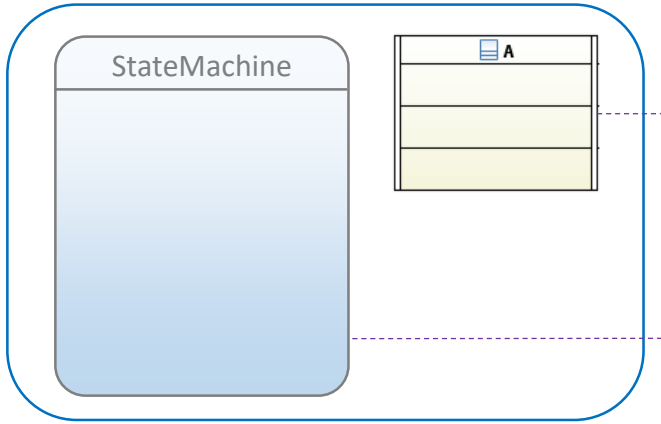
Map notation symbols to syntactical elements

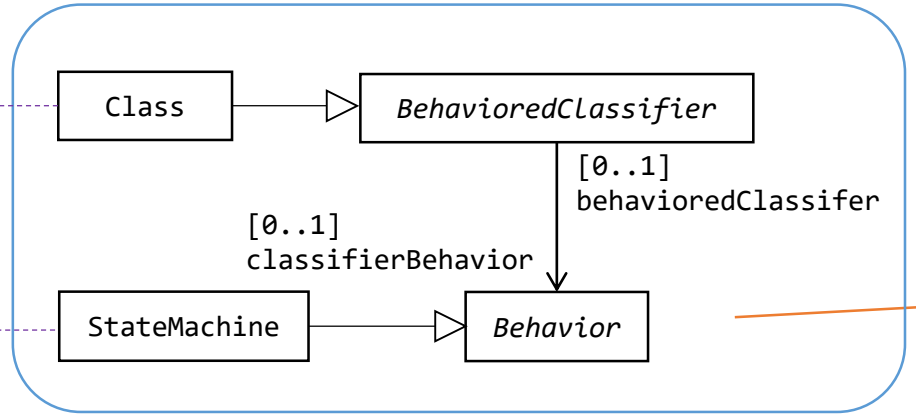
Subject ⇨ Verb ⇨ Complement

Plays role of ↑ ↑ Is a ↑ Plays role of

我是学生

Wǒ shì xué shēng



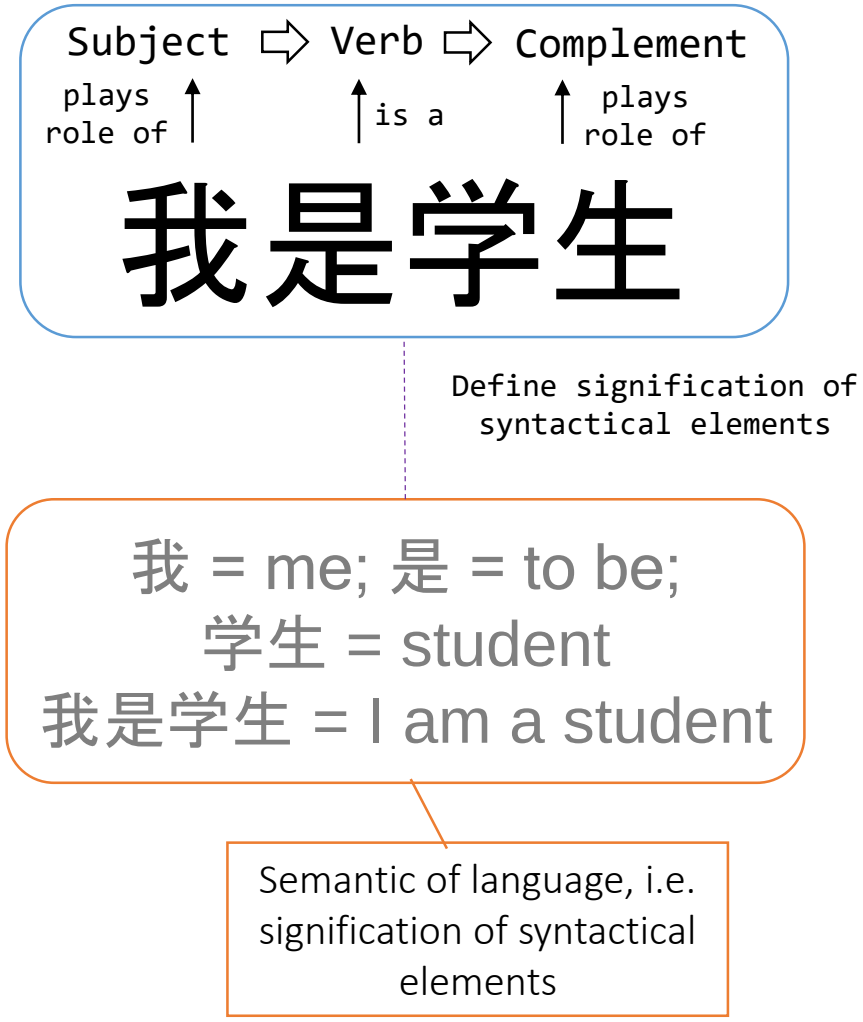


Note: some concepts in a language may not have a representation.

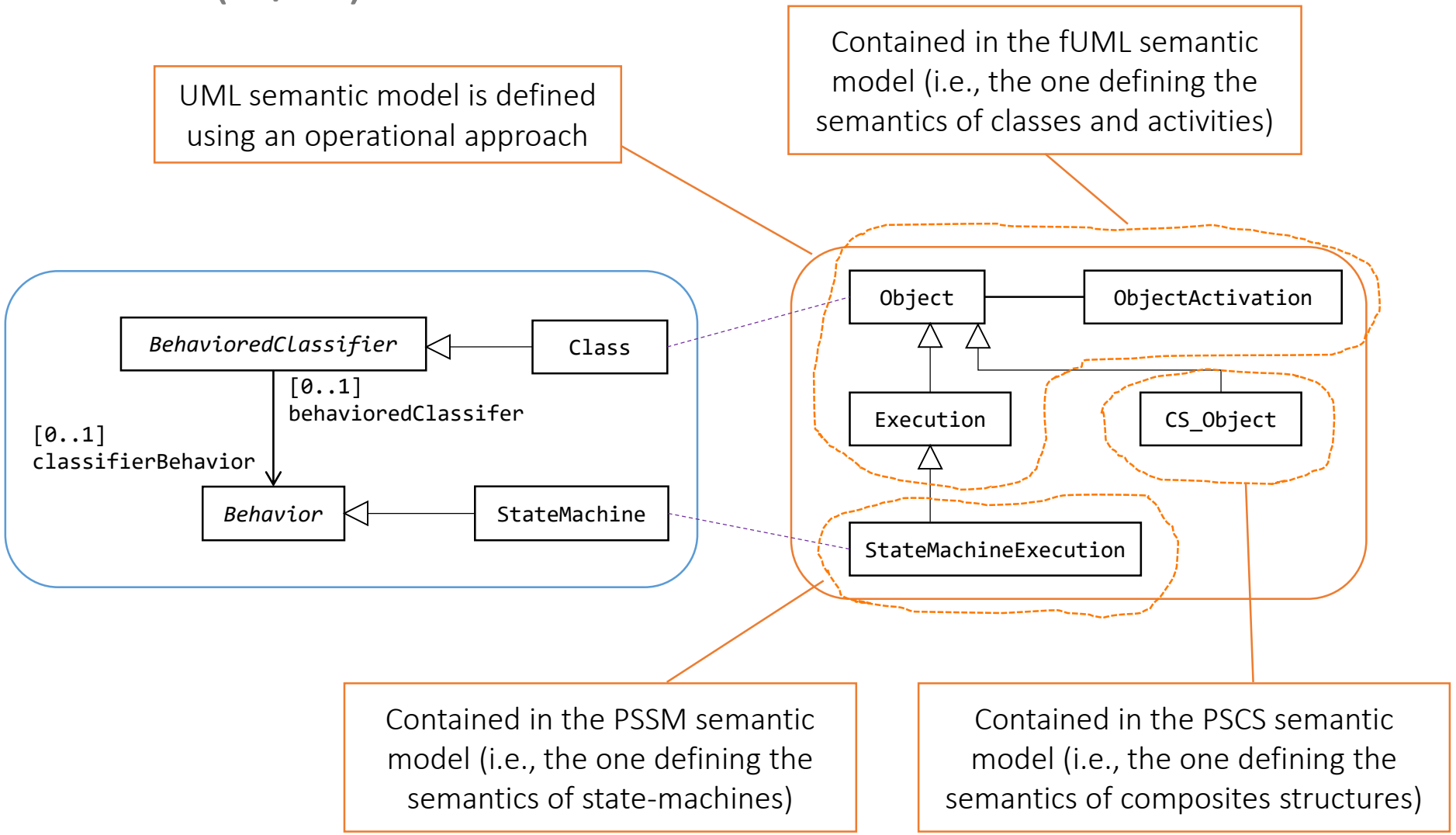
The meaning of the concept defined by a language semantics (1/2)

The semantics of a languages in computer science

- Defined by a semantic model
  - Captures the signification of concepts provided by the syntactical model
- Semantic model can be formalized using different approaches
  - Operational – semantics is defined by a program / algorithm
  - Axiomatic – semantics is defined as a set of axioms)
  - Denotational – semantics is defined as a set of denotations
  - Transformational – semantics is the one defined by the target language
- The form of the semantics defines what you can do with it
  - Execution/reasoning about valid execution traces for a model…



The meaning of the concept defined by a language semantics (2/2)



Abstract syntax, Notation, and semantics WRAP UP

- The abstract syntax
  - The concepts that are offered by the language
  - Papyrus: you use it when you create UML elements
- The notation
  - The representation that is attached to language concepts
  - Papyrus: you use it when creating diagrams and graphical elements
- The semantics
  - The elements capturing the meaning of language concepts
  - Papyrus: you use it when you execute a model

Domain specific modeling languages (DSLs)

1. **Recap – abstract syntax, notation, and semantics**
2. **What are domain specific languages, what are their objectives?**
3. **Text-based DSLs**
4. **UML profile mechanisms**
5. **Define a simple profile**

What are domain specific languages (DSLs)?

- **A language specialized to a particular application domain (contrast to a general-purpose language (GPL), which is broadly applicable across domains).**
- **Wide variety of DSLs, ranging from the quite generic HTML to very specific ones**
- **Can be further subdivided into specification/modeling languages and domain-specific programming languages.**
- **Special-purpose computer languages not new, term "domain-specific language" has become more popular due to the rise of domain-specific modeling.**
- **Separation between general-purpose and domain-specific not always sharp**
  - **Language may have features for particular domain but applicable more broadly**
  - **Conversely capable of broad application but in practice used primarily for specific domain**
  - **Examples: Perl and PostScript**

Objectives of DSLs

1. **Use domain specific concepts and vocabular**
2. **Small, just the concepts that are required**

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Textual modeling languages

- Classical context free grammars

- Xtext (Eclipse)

Context free grammars (CFGs)

- Grammar = set production rules describing all possible strings of a formal language.
- Production rules iof n the form A → α
- A – nonterminal symbol, α – string of terminals or non-terminals
- “context free” – production rules can be applied regardless of the symbols before/after.
- Example : Statement → Id = Expr ;

Terminal

Classical tools

- Lex and yacc (free / enhanced variants flex and bison)
- Two steps
  - Lexical analysis
  - Syntactical analysis (parsing)

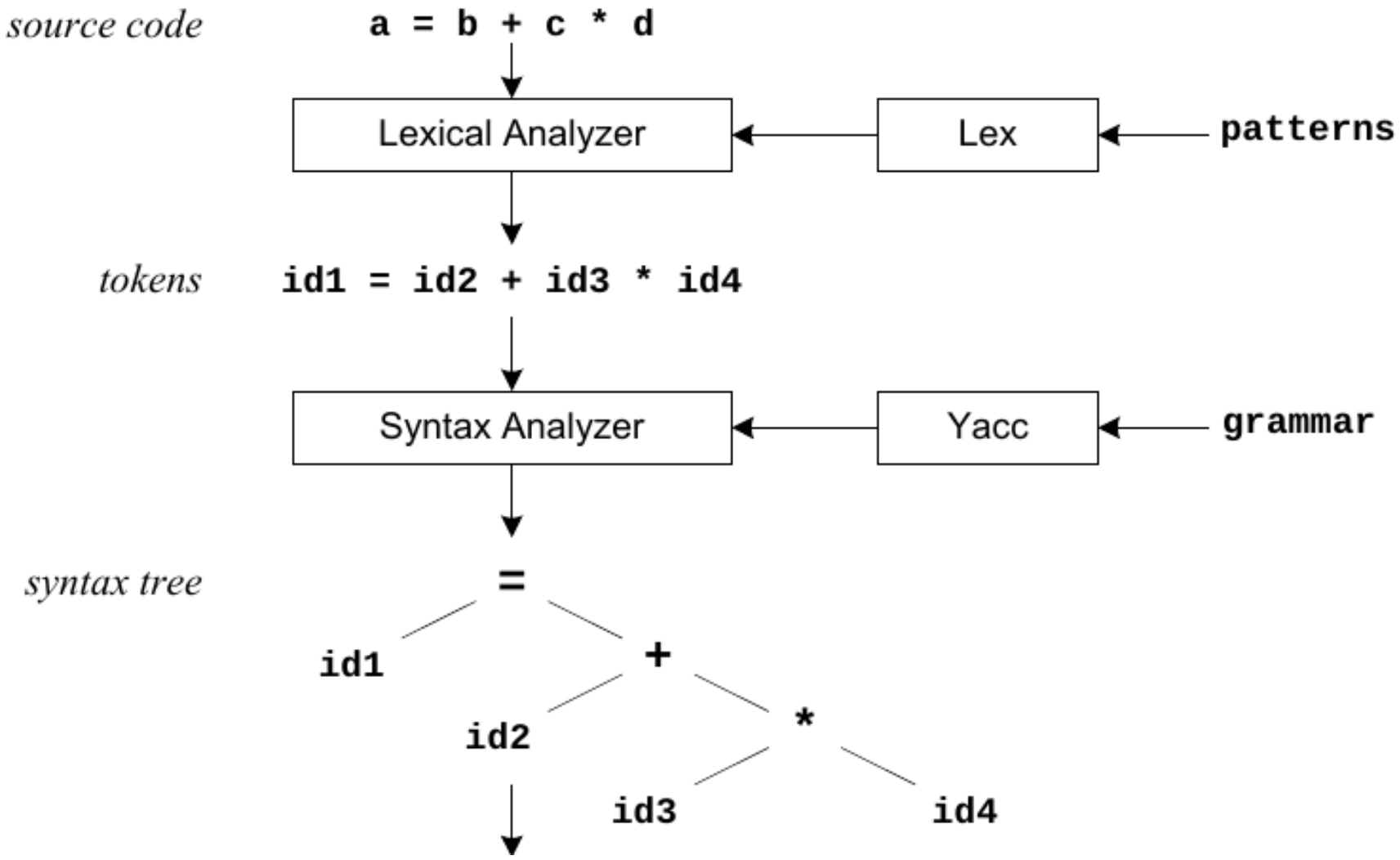
Regular epressions

Meta character	Matches
.	any character except newline
*	zero or more copies of the preceding expression
+	one or more copies of the preceding expression
?	zero or one copy of the preceding expression
a|b	a or b
[]	character class

[A-Za-z0-9]+ at least one alpha numerical character

[A-Za-z][A-Za-z0-9_]* start with a alphabetic character, followed by zero or more alpha numerical characters (+ underscore)

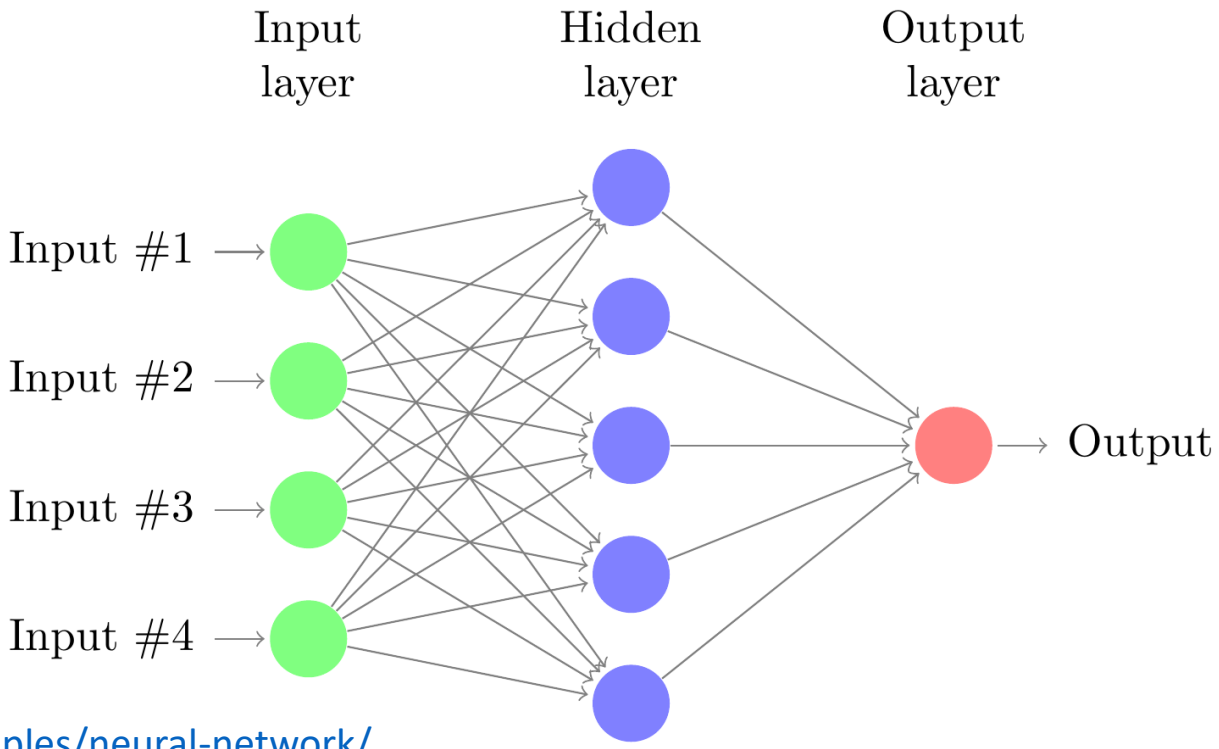
Example


source code
a = b + c * d
Lexical Analyzer
Lex
patterns
tokens
id1 = id2 + id3 * id4
Syntax Analyzer
Yacc
grammar
syntax tree
=
id1
+
id2
*
id3
id4

Other example – neural network descripton

- Keras https://keras.io/
- Python library/framework for describing and executing neural networks
- High level API

- Goal : Declarative description of neural network



Source: https://texample.net/tikz/examples/neural-network/

Neural network descripton

- Different layer types in Keras
- Each layer has a set of parameters
  - Simplification – focus on two layers and a subset of the parameters
- Convolutional layer Conv1D
  - filters: Integer, dimensionality of output space (i.e. number of output filters in the convolution).
  - kernel_size: An integer specifying the length of 1D convolution window.
- LSTM layer (Long Short-Term Memory layer – recurrent layer)
  - activation: Activation function to use.
    Two options: **tanh** - hyperbolic tangent, **linear** - a(x) = x.
  - use_bias: Boolean (default True), whether the layer uses a bias vector.

Other example (from neural networks)

Possible textual description

```
network MyNN {
        l1 : Conv (kernel_size = 2)
        l2 : LSTM (use_bias = true, activation = tanh)
        l3 : LSTM (activation = linear)
}
```

Howto describe this editor formally?

Use Eclipse xtext

- https://www.eclipse.org/Xtext/

"Xtext is a framework for development of programming languages and **domain-specific languages**. ... define your language using a powerful grammar language. … get parser, linker, typechecker, compiler as well as editing support for Eclipse, any editor that supports the Language Server Protocol and your favorite web browser."

Xtext formalization

```
grammar org.xtext.example.mydsl.NNDsl with org.eclipse.xtext.common.Terminals

generate NNDsl "http://www.xtext.org/example/nndsl/NNDsl"

Model:
    'network' name=ID '{'
        (layers += Layer)*
    '}';

Layer:
    name=ID ':' (
        'Conv' options += COptions |
        'LSTM' options += LOptions
    );
…
```

(see in Eclipse editor, exercise)

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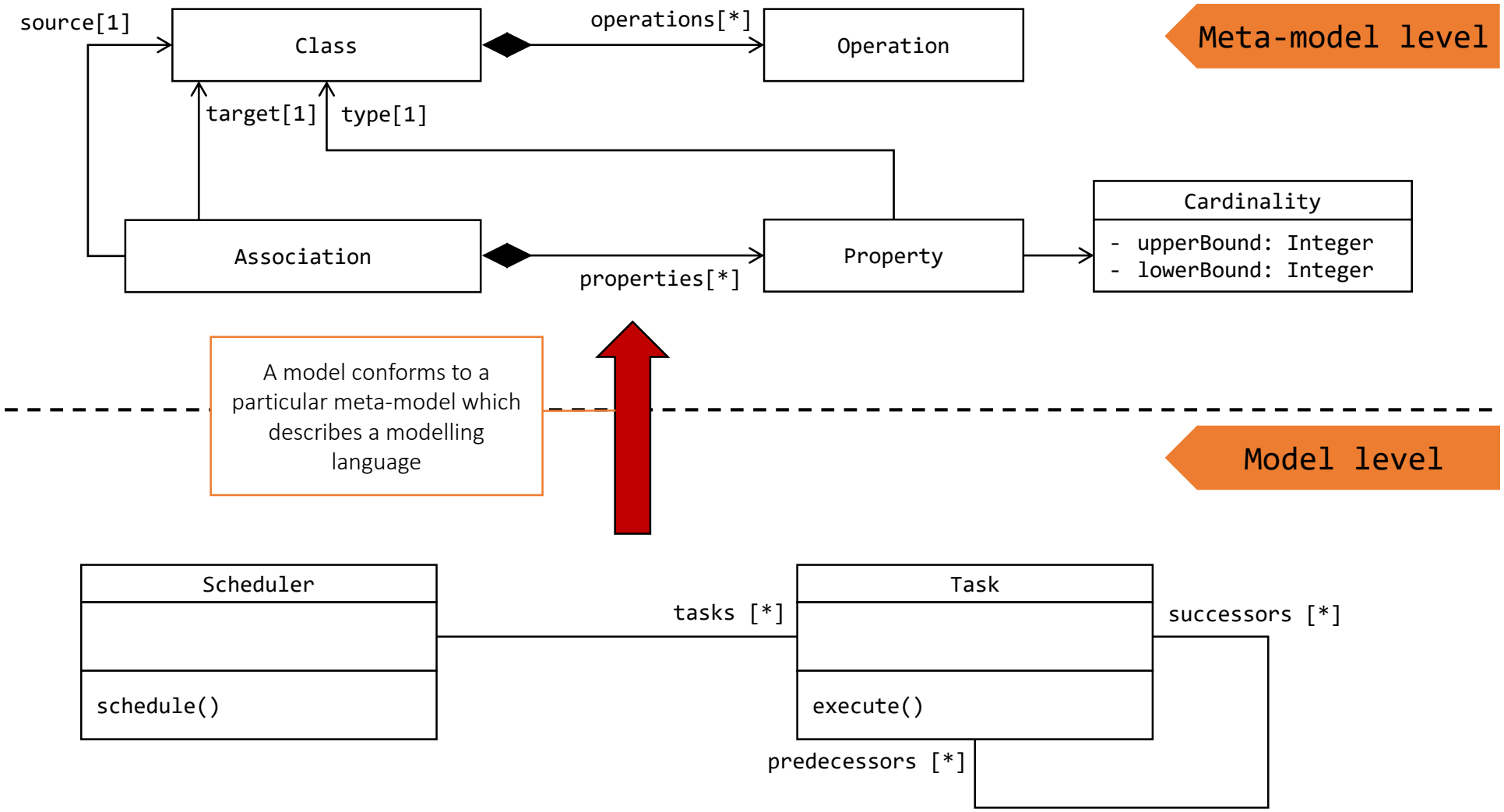
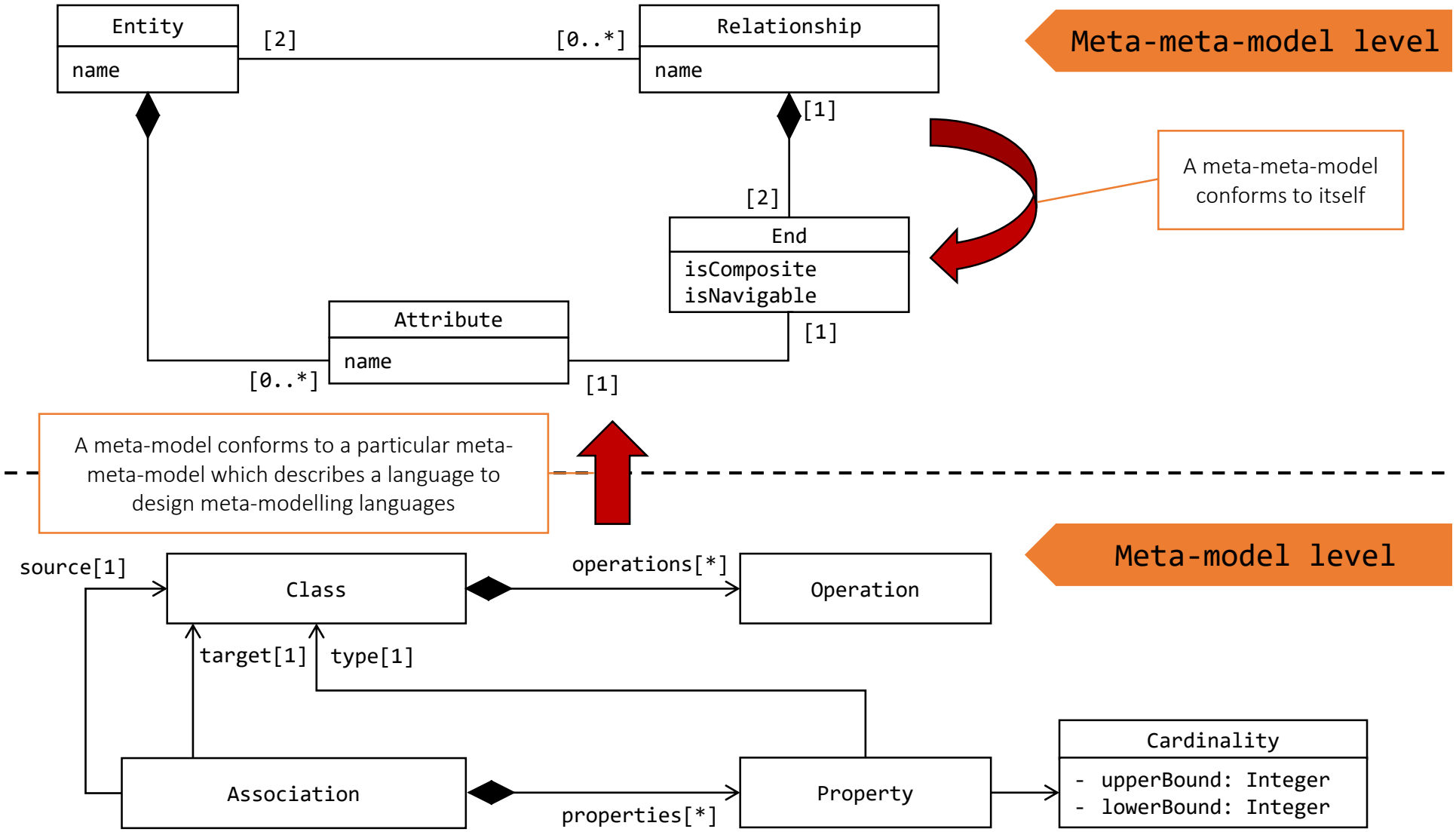
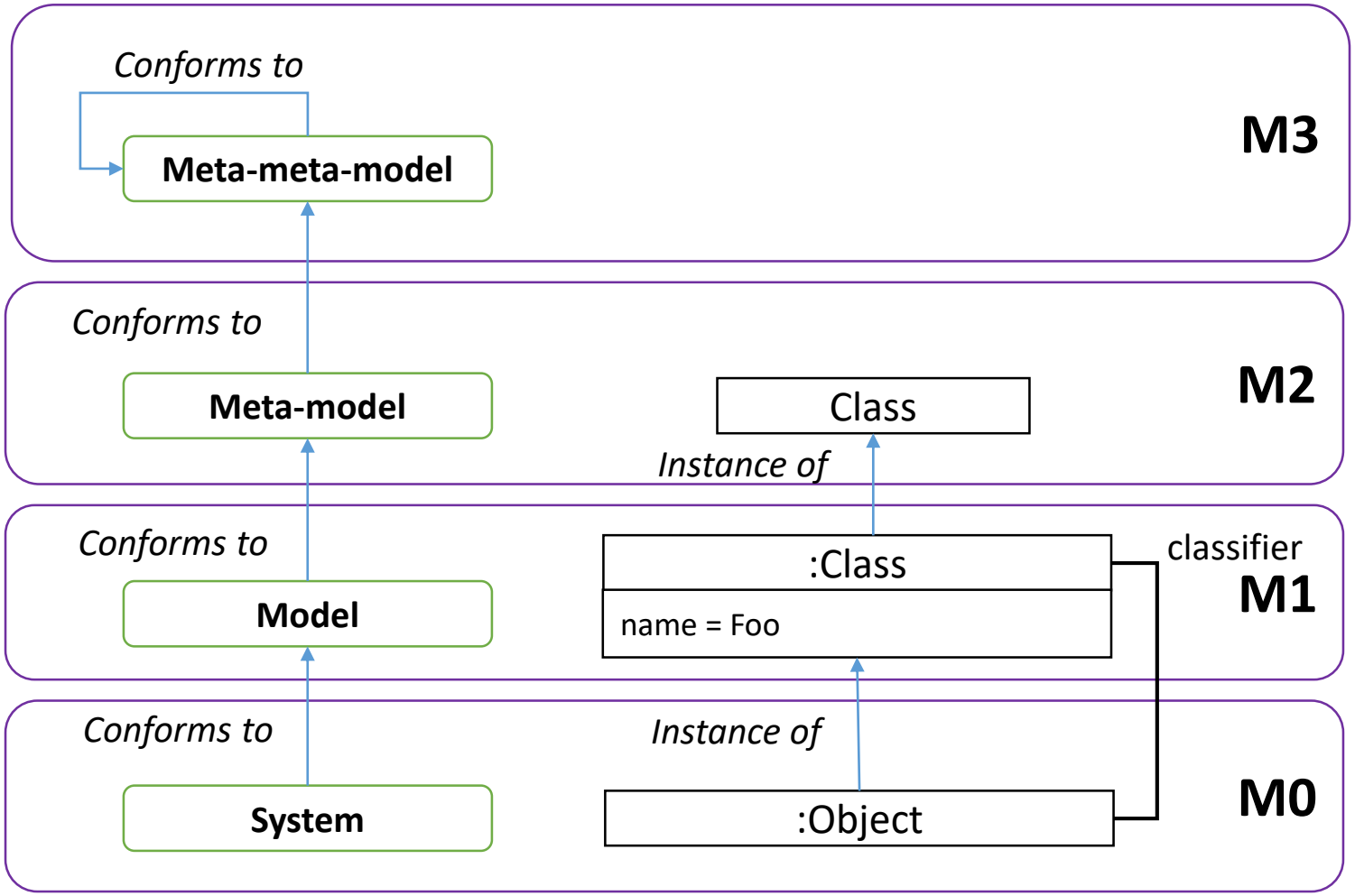
Model and metamodel – conformance relationship



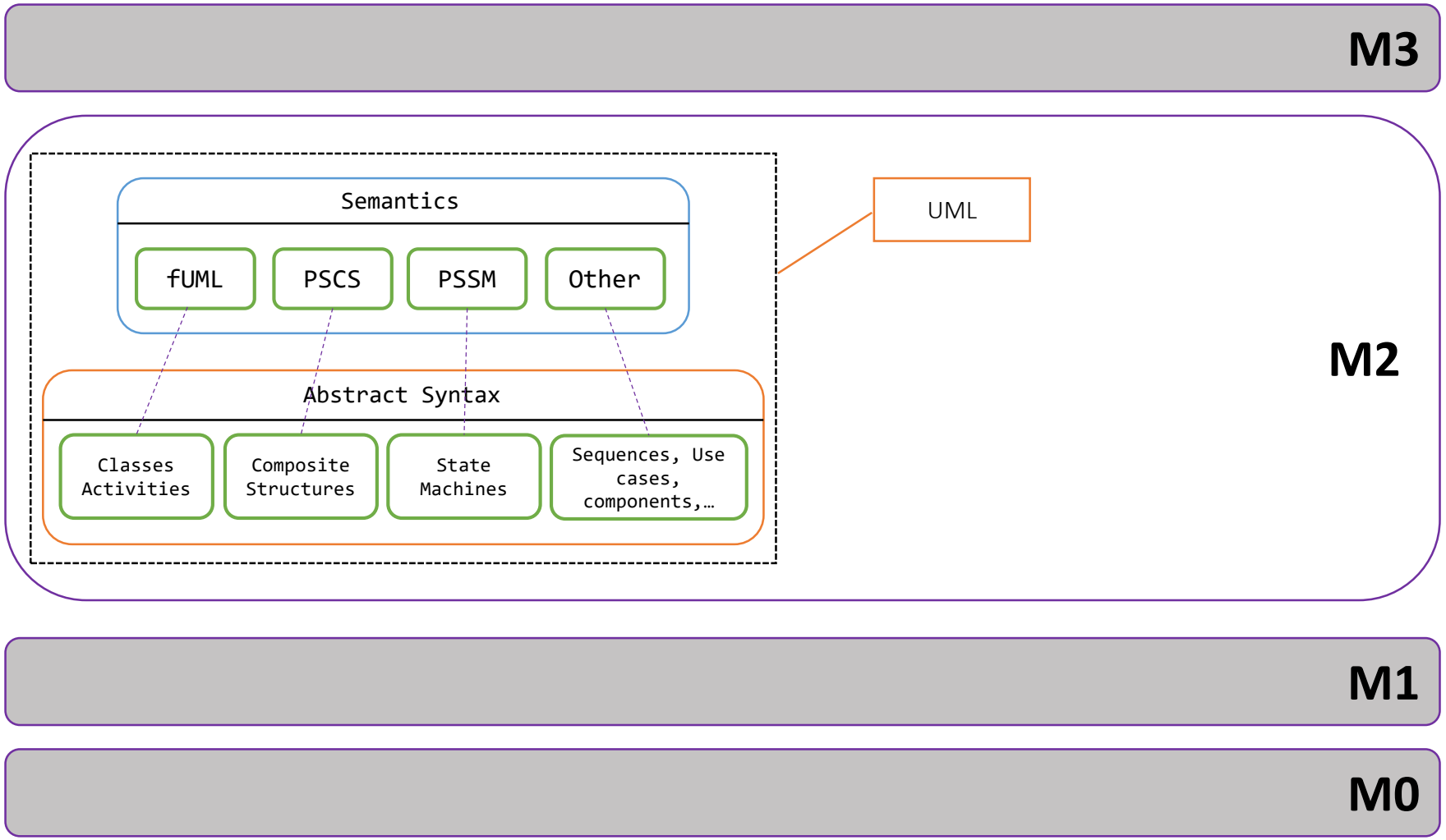
Metamodel and meta-meta-model – end of the loop



Wrap UP - Model, meta-model and meta-meta-model



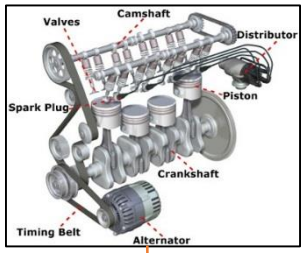
WHERE IS UML ?


M3
Semantics
fUML
PSCS
PSSM
Other
UML
Abstract Syntax
Classes Activities
Composite Structures
State Machines
Sequences, Use cases, components,…
M2
M1
M0

UML – richness & limitations

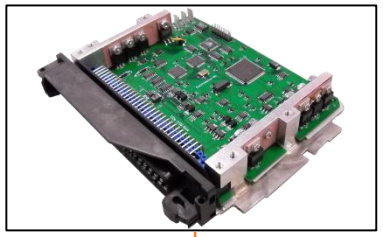
- Formalize structure of a system using different formalisms (statics)
  Formalize behavior of a system using different formalisms (dynamics)
  Capture system abstractions and user interactions
- Intended to be agnostic of a particular domain
  - But it is clearly software oriented
  - Many concepts come from object oriented programming languages
- Will you use it as it is to describe the following kind of systems ?

© http://www.driving-test-success.com



Points of interest: piston diameter, cylinder volume, weight of components, etc.

© http://www.standaloneenginemanagement.com/

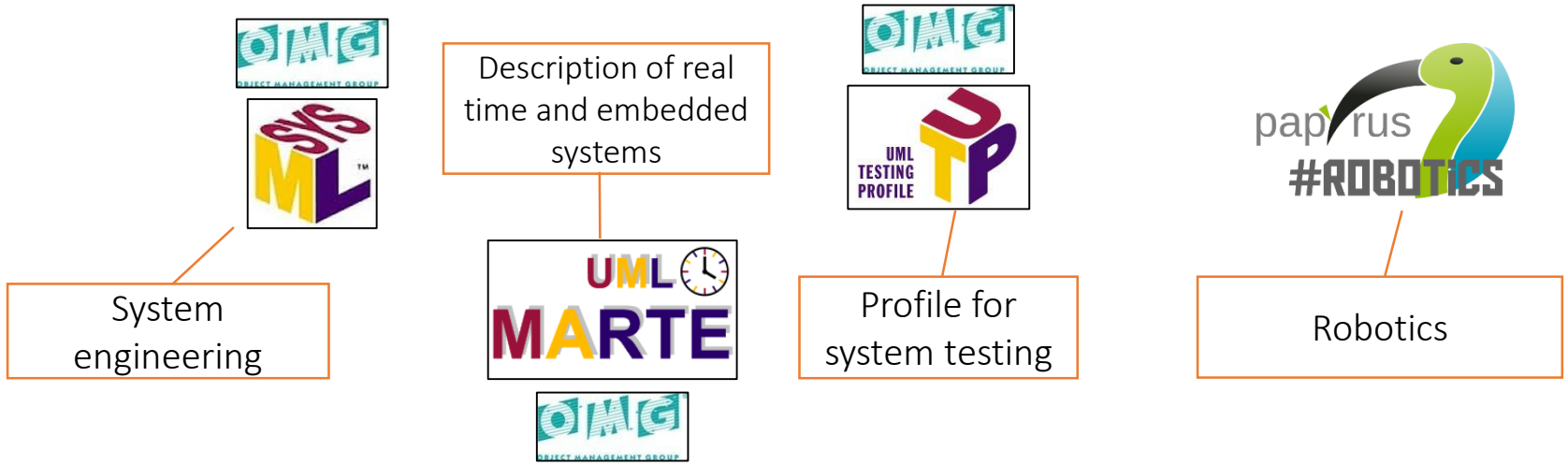


Points of interest: processor rate, the quantity of memory, energy consumption?

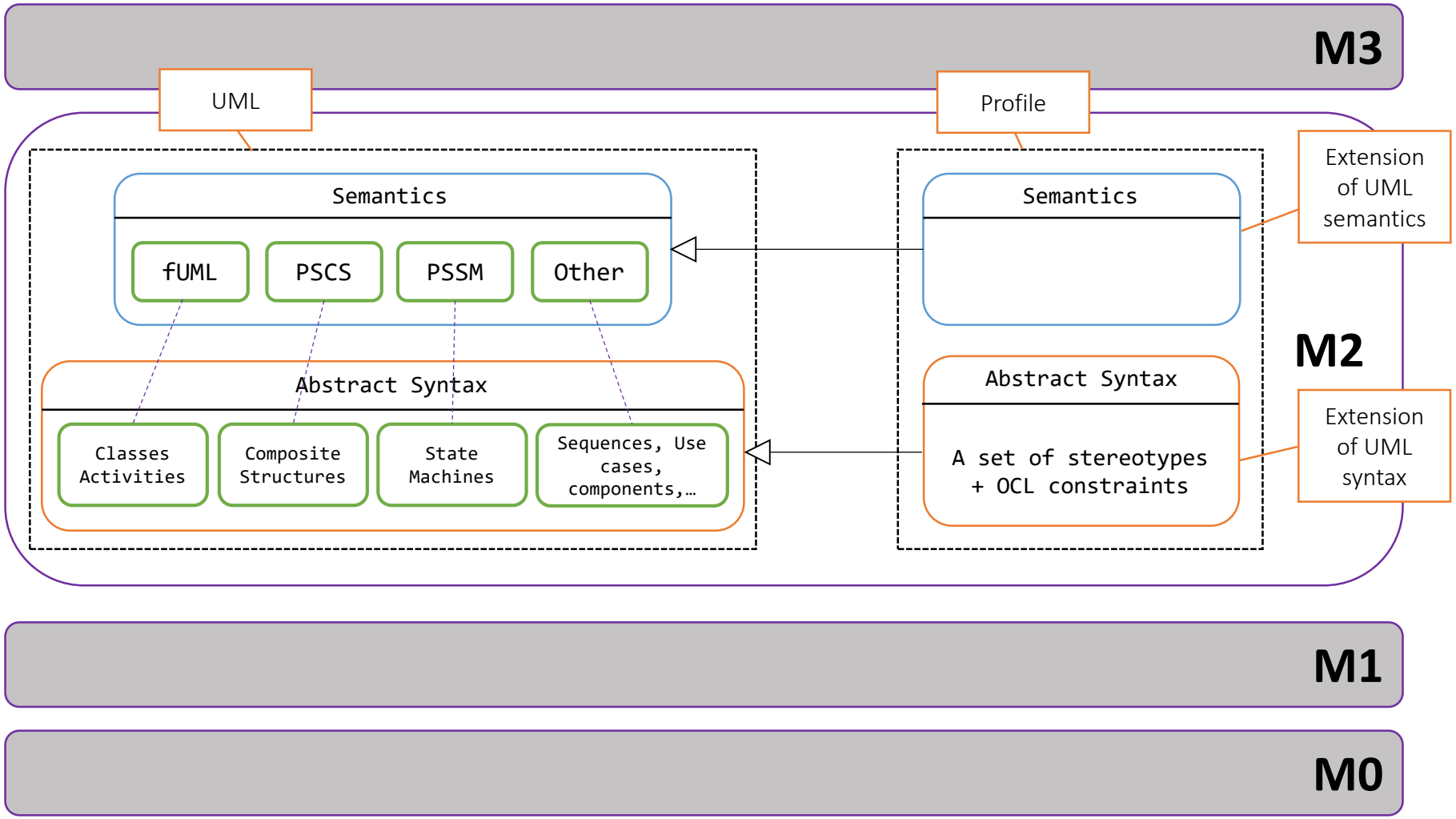
UML extensibility – the profile mechanism

Difficulty to capture concerns of specific domains

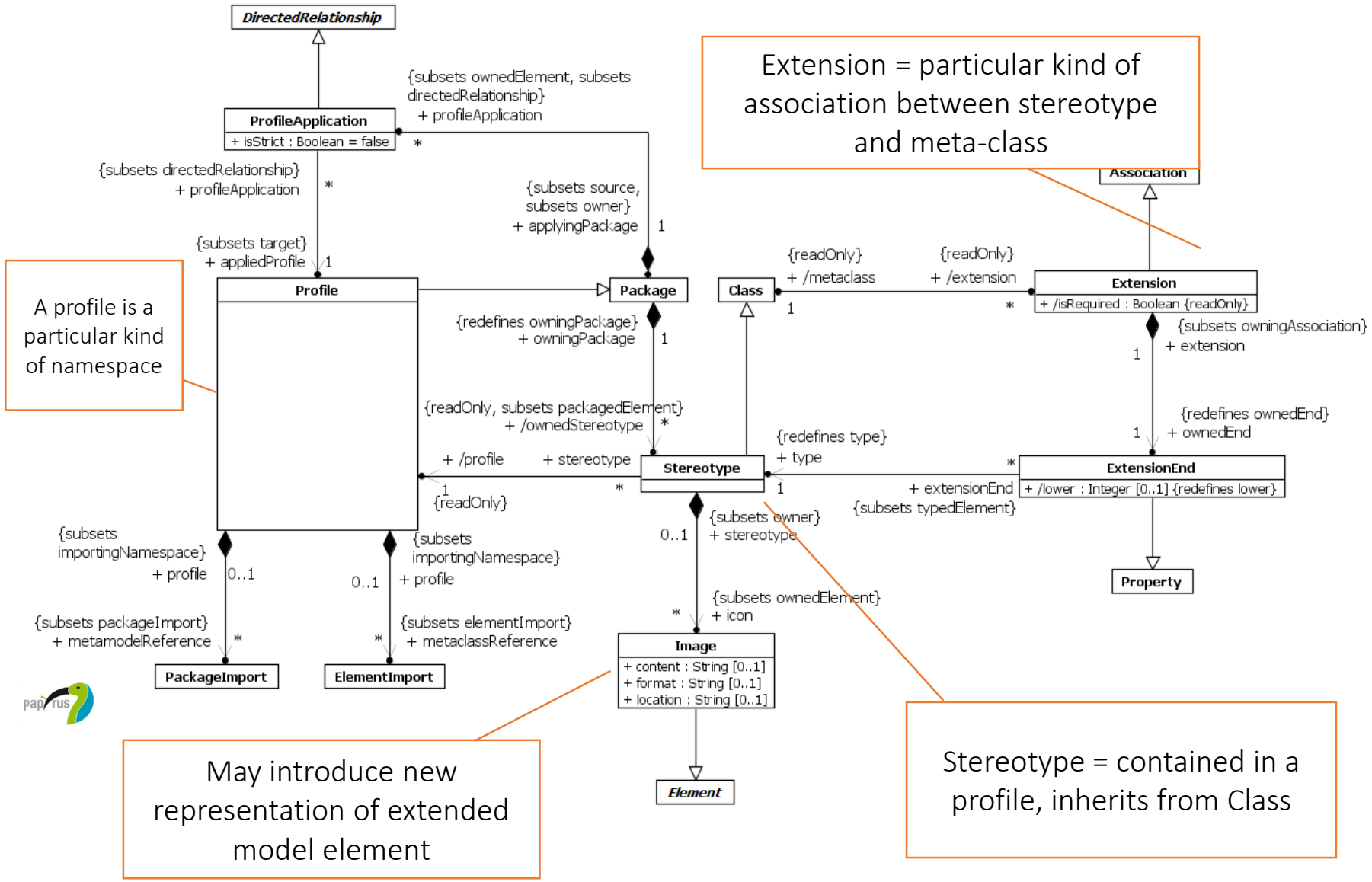
- ….such as mechanics, electronics, avionic, etc.
- Extend UML to capture the concerns of these domains
  - Enable design of UML based domain specific languages
  - Capitalize on syntax and semantics provided by UML
- UML profiles are widely used in industry



How does a profile contribute to UML?



Profiles - UML meta-model view



Extension = particular kind of association between stereotype and meta-class

A profile is a particular kind of namespace

May introduce new representation of extended model element

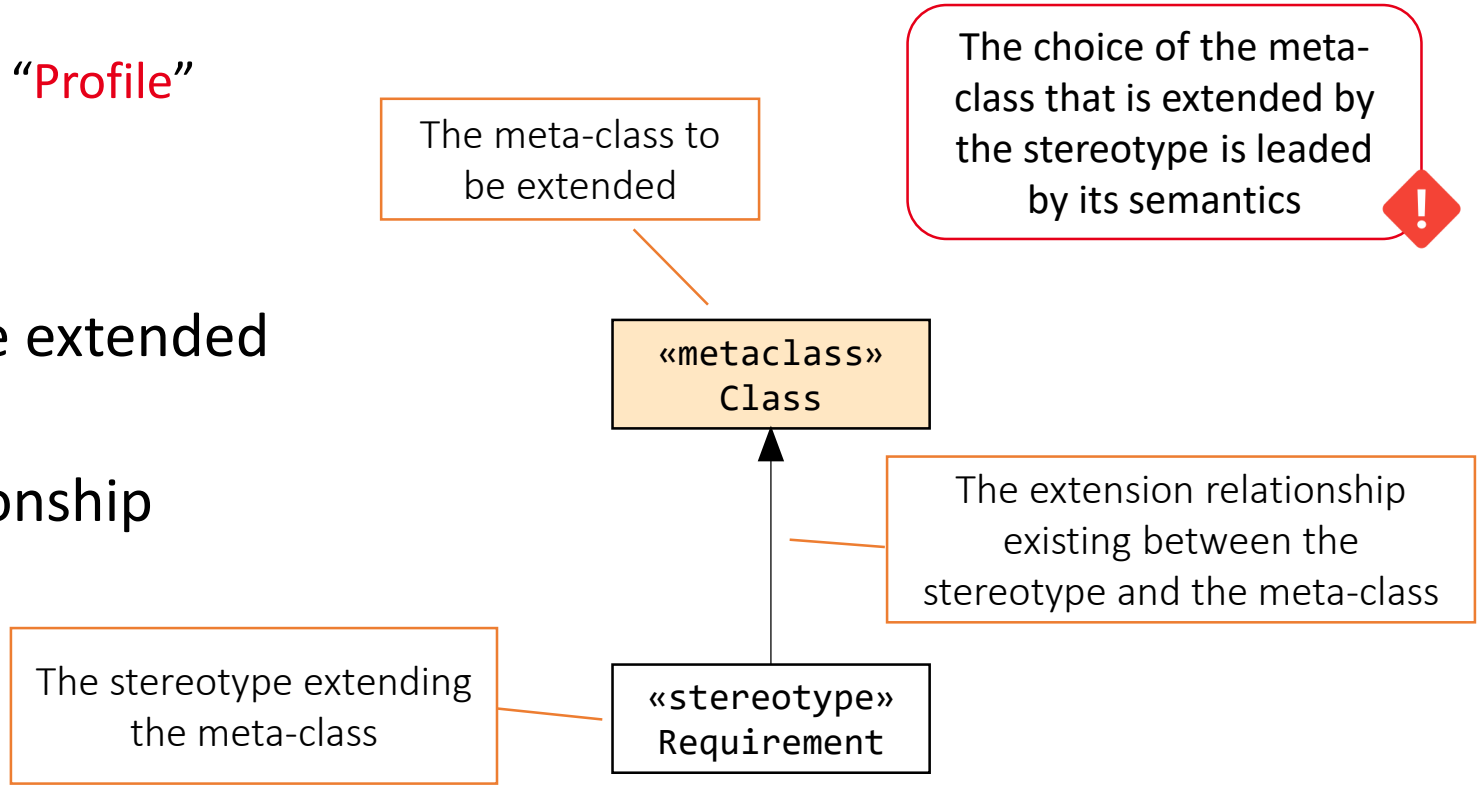
Stereotype = contained in a profile, inherits from Class

Example – a profile for requirement engineering

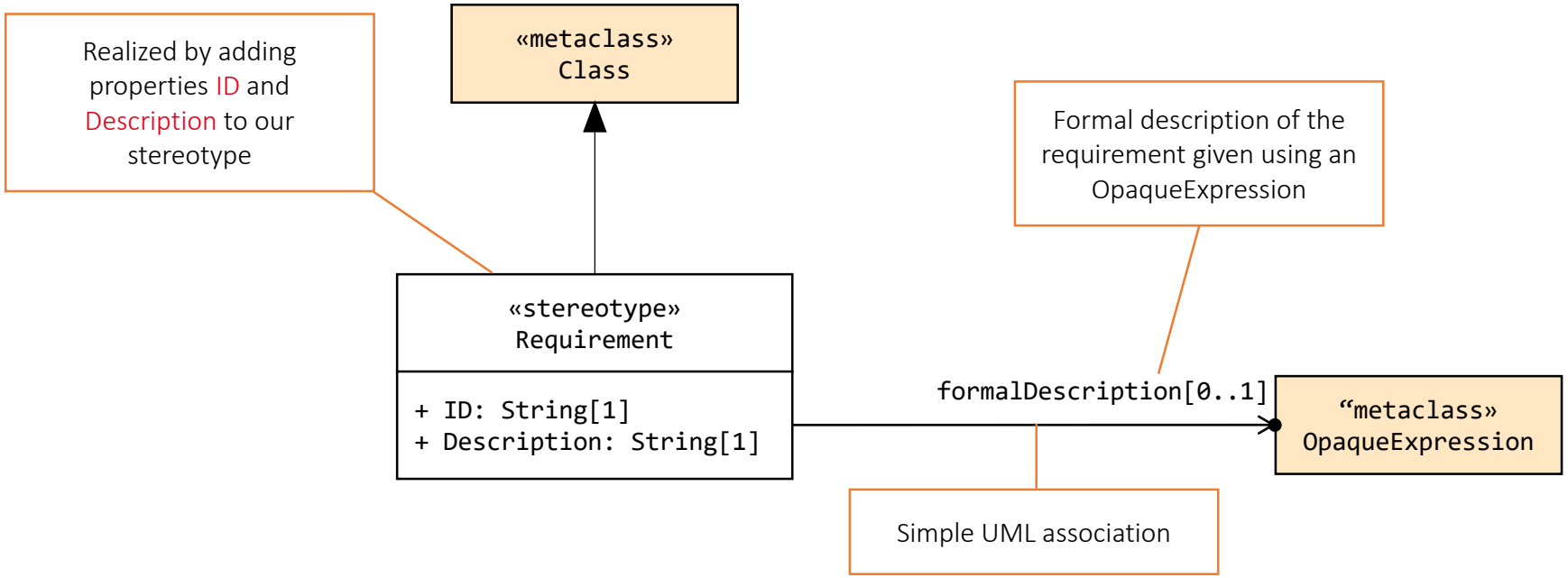
- Objective – Capture system requirements
- Analysis
  - In my system I can have
    - Functional requirements (actions that can be accomplished by the system)
    - Performance requirements (expected performance for the system)
    - Structural requirements (Identify the necessary structure of the system)
  - A base requirements has
    - An identifier
    - A text that describes a requirement
    - A formalization (i.e. an expression given in a formal language)
  - A general requirement can be refined (clarified) by a set of sub-requirements

Step 1 – define a «Requirement» stereotype

- In Papyrus
  - File – New – Papyrus project
    - Set the project name
    - Choose architecture context “Profile”
- Start to define your profile
  - Import the meta-class to be extended
  - Create a stereotype
  - Create the extension relationship



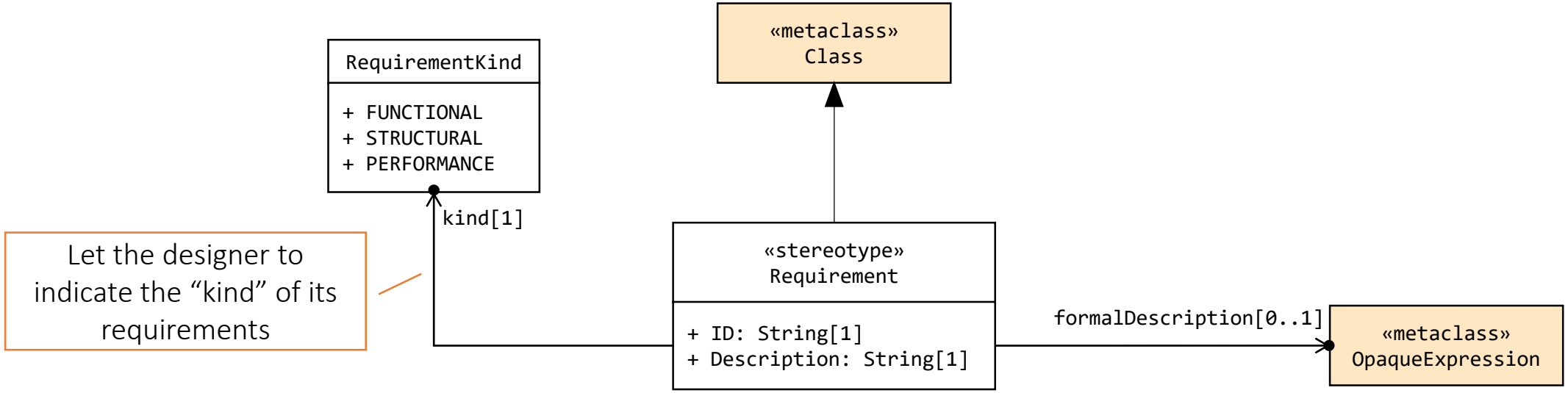
Step 2 – capture basic properties of a requirement



Why did we choose OpaqueExpression?

- Possibility to choose a formal language to define something
- The specification takes the form of a piece of text that is easily accessible

Step 2 – capture basic properties of a requirement



Question

- What is missing in the profile to make a requirement "refinable" by another requirement?

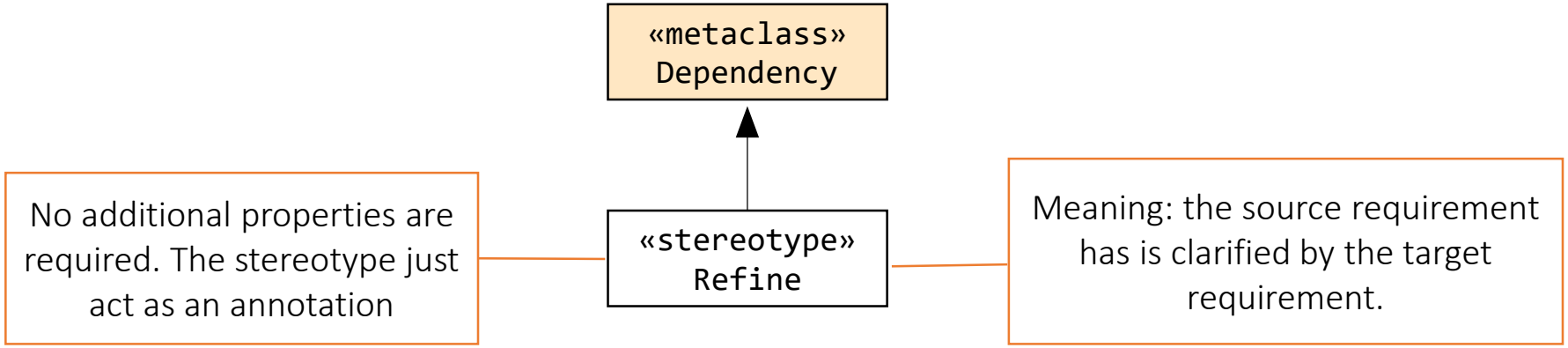
Step 3 – Refinement relationship

Objective

- Find meta-class that allows the specification of a relationship between two classes
- Create a stereotype that apply on this meta-class

Proposal

- Dependency (UML 2.5 – p. 36)
  - Source and target are NamedElements
  - A model element requires another model element for its specification



Reduce UML expressiveness – Why and How

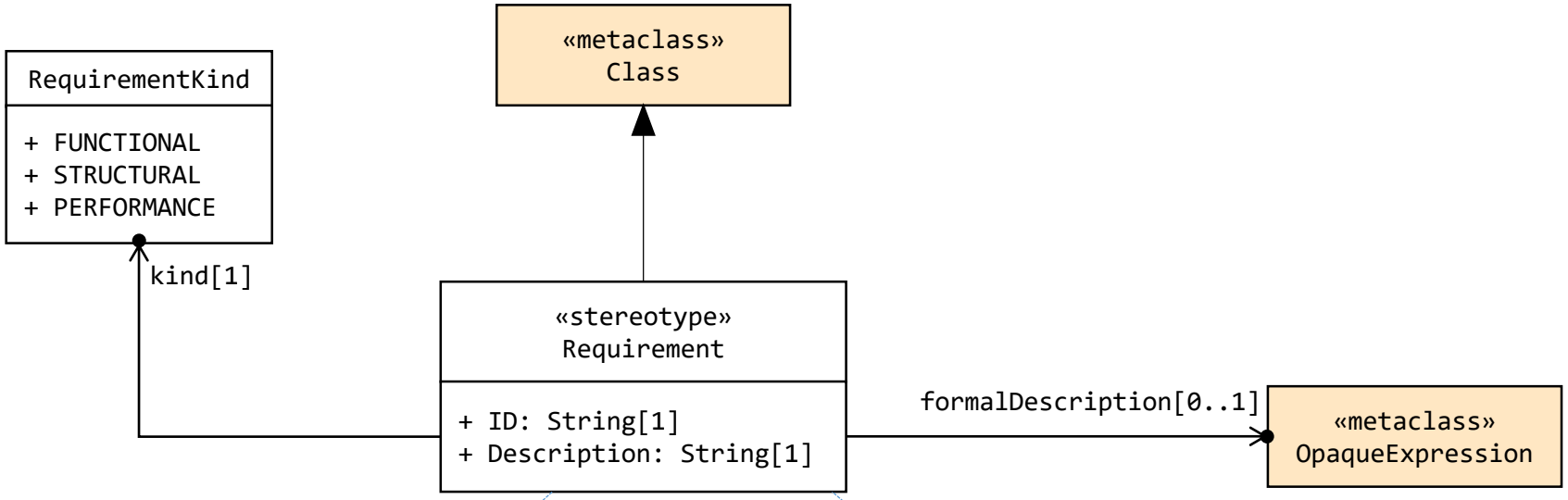
Why? We do not need some features when specifying requirements

- Class
  - ~~A class can have operations~~
  - ~~A class can have a classifier behavior~~
  - ~~A class can be active~~
- Dependency
  - ~~The source and the target can be NamedElement~~

How? Via OCL constraints in our profile

- Object Constraint Language
- OCL is a formal language
- Specification of constraints on models (not only UML)
- The constraints that can be verified at any time of the model life

Reduce expressiveness – «Requirement» stereotype

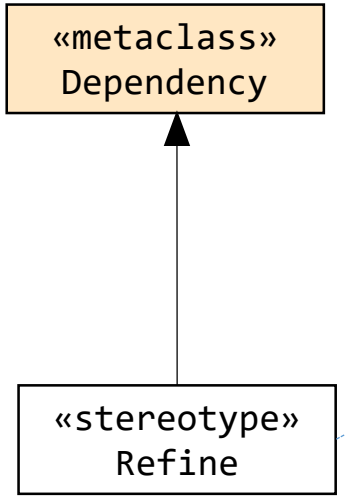



```
self.base_Class.getOwnedOperations()->isEmpty()
                                  and
self.base_Class.classifierBehavior->oclIsUndefined()
```

{}

```
self.formalDescription.language->size()=1
                                  and
self.formalDescription.language->first()="OCL"
```

Reduce expressiveness – «Refine» stereotype

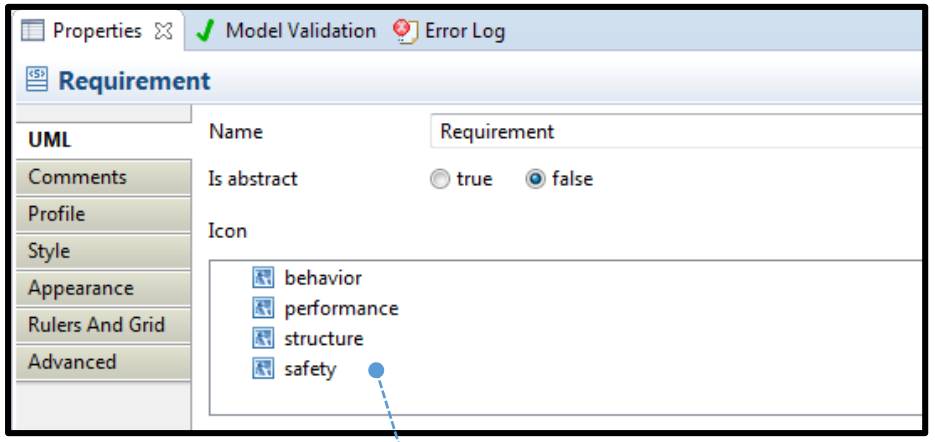


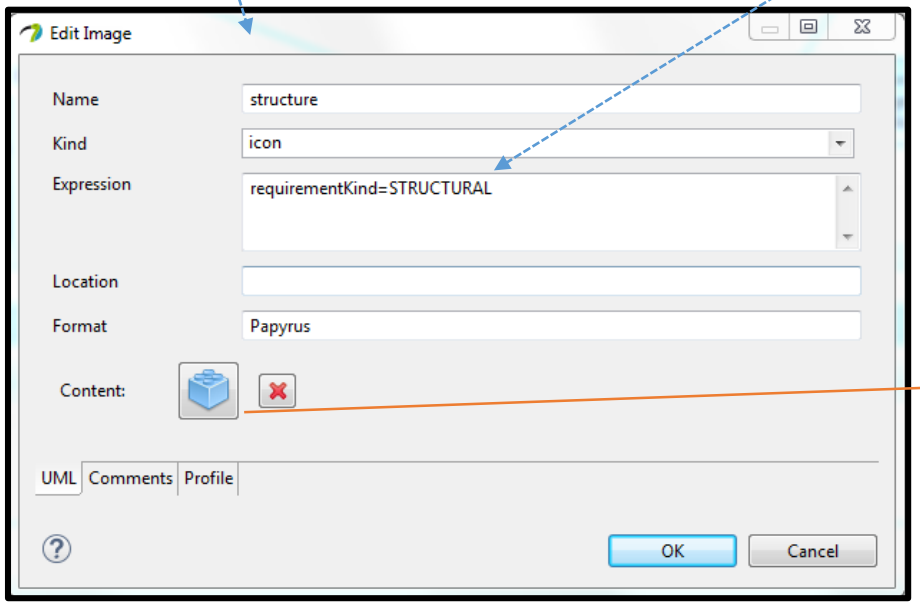

```
self.base_Dependency.supplier->
        getAppliedStereotype("RequirementProfile::Requirement")<>null
                                    and
self.base_Dependency.client->
        getAppliedStereotype("RequirementProfile::Requirement")<>null
```

Guarantee that the source and the target are requirements

- Help to maintain a model correct by construction

Customize stereotyped model elements





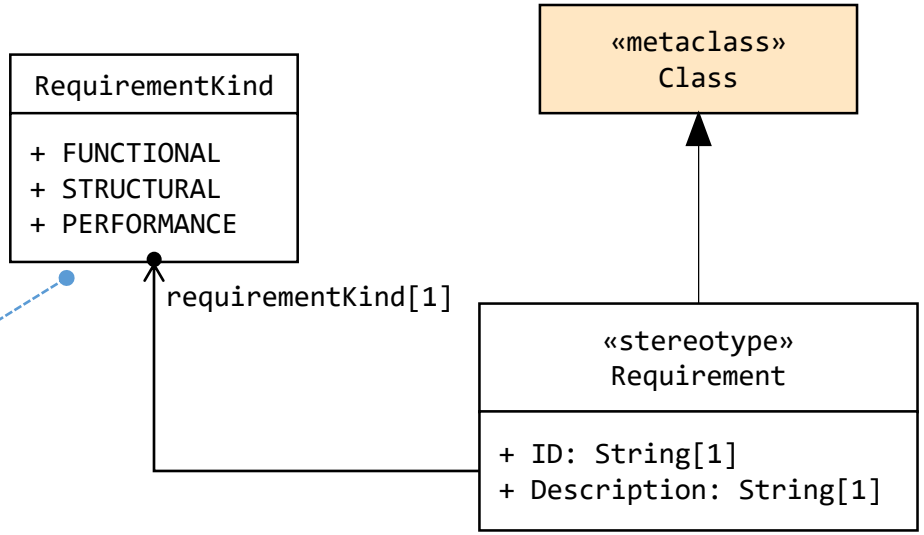
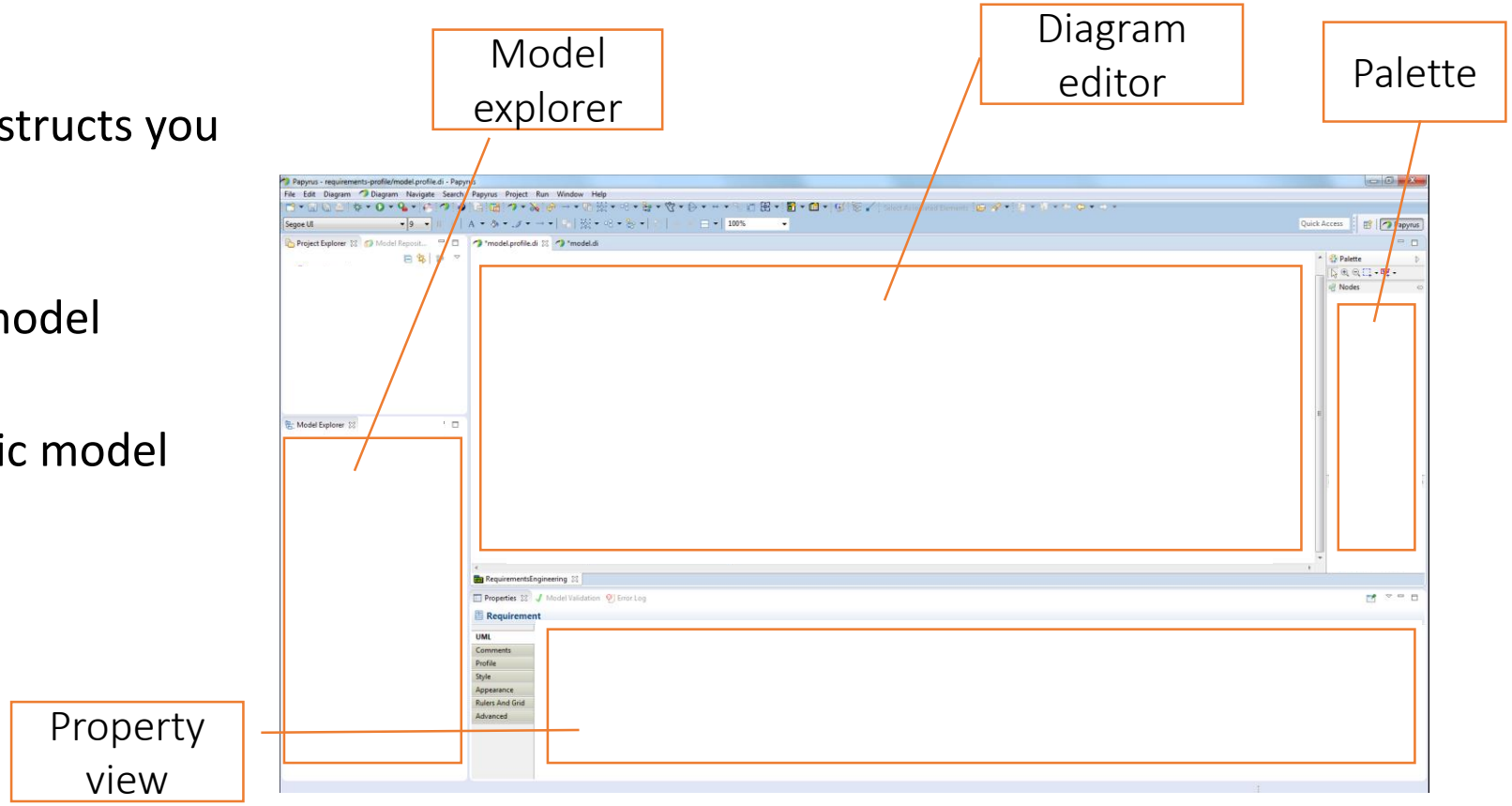


Image associated with representation when
(1) stereotype «Requirement» is applied and
(2) its "requirementKind" property = "STRUCTURAL"

[limitation – bitmap icons, use CSS for SVG shapes]

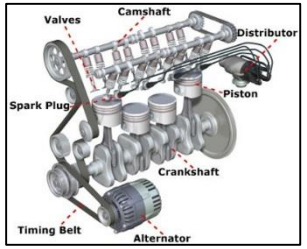
Customization – Papyrus tooling

- **Objective** – a modeler focused on a particular domain … based on a UML profile
- Customizable elements
  - **Palette**
    Provide the modeling constructs you can use in diagrams
  - **Model explorer**
    Current structure of the model
  - **Property view**
    Edit properties of a specific model element
  - **Diagram editor**



Apply a profile => tailored version of UML

After applying the profile on a model, we can capture requirements

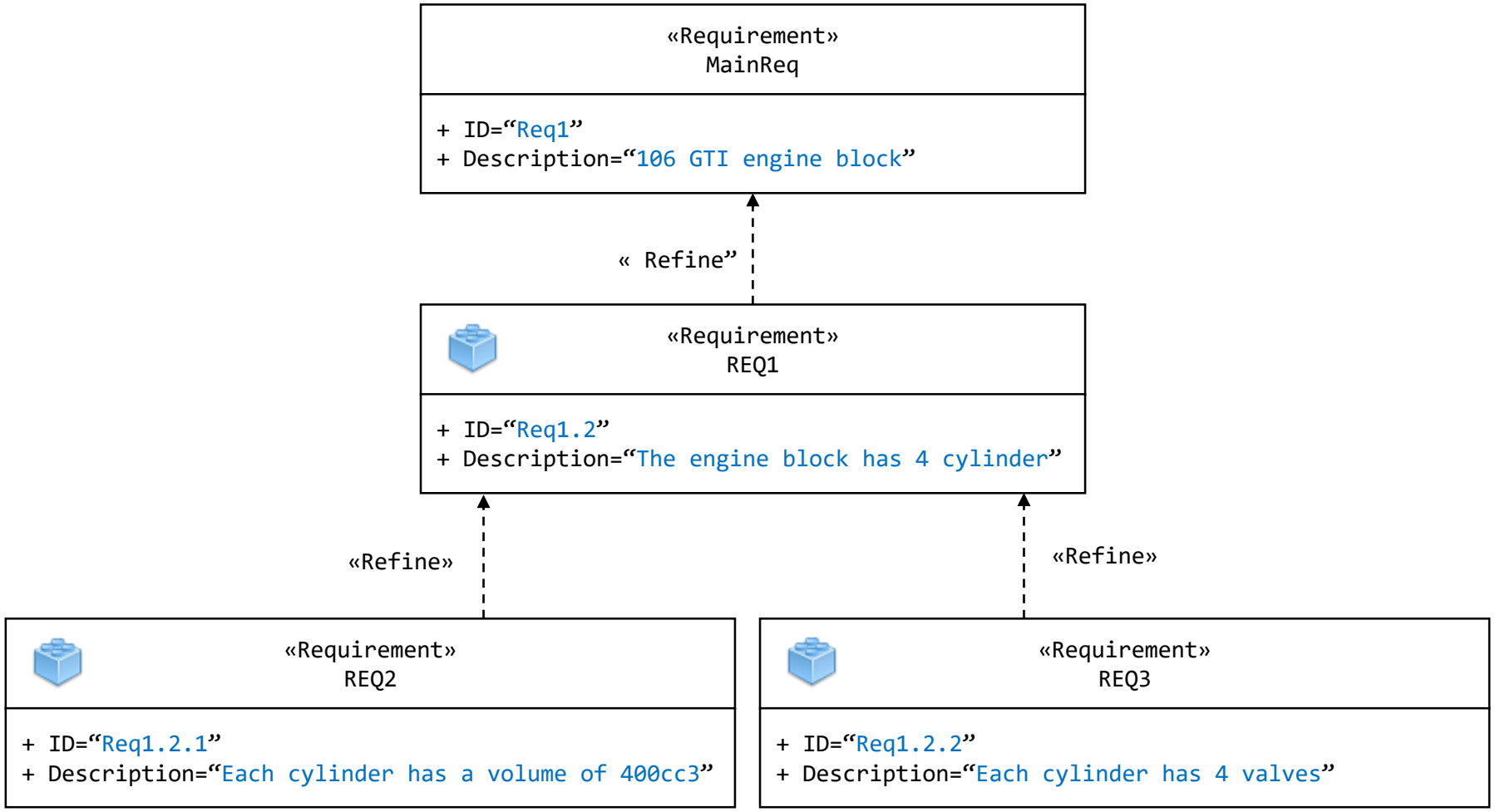


Lets consider the following specification

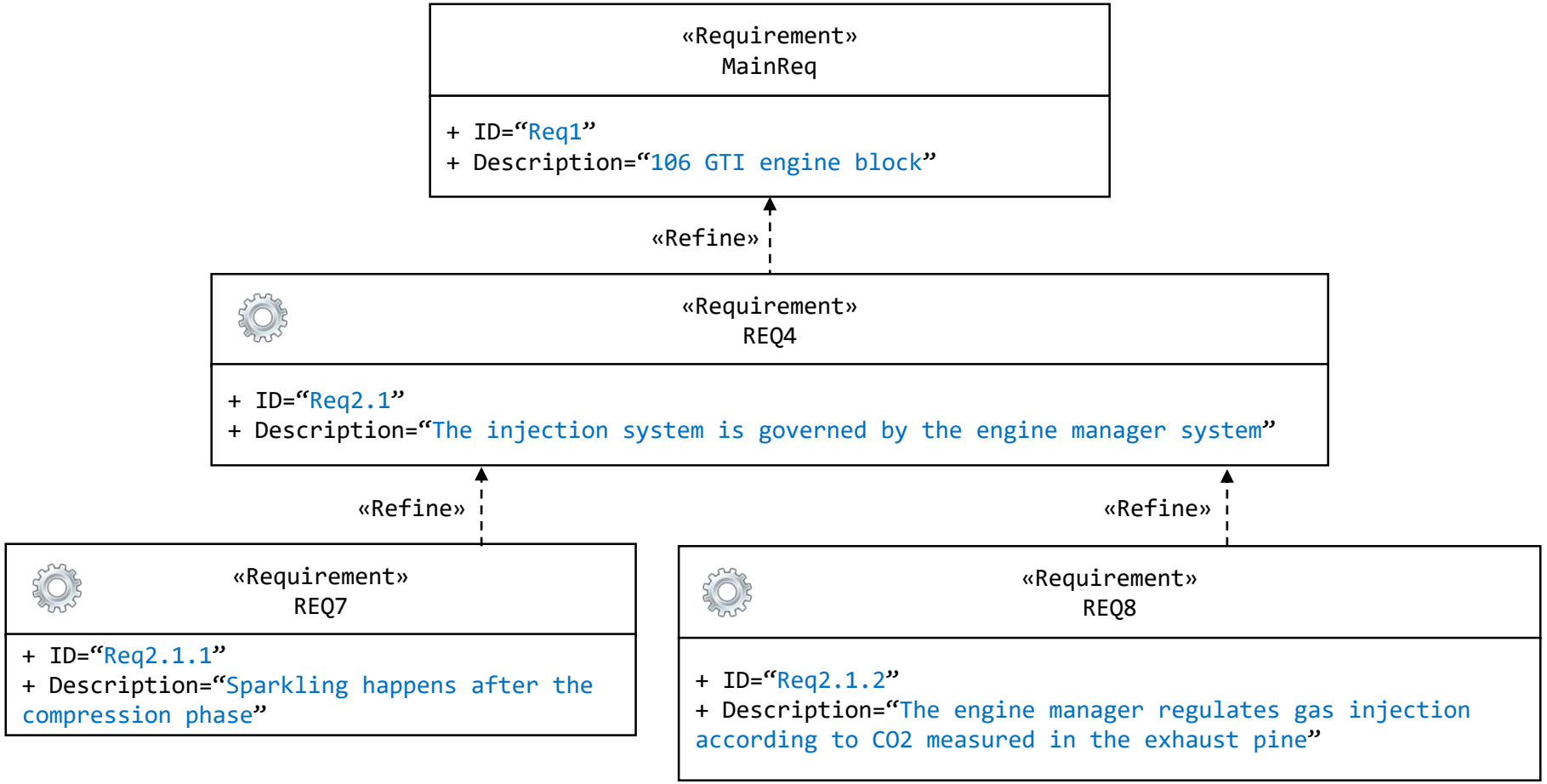
- REQ1 - The engine block of a 106 GTI has four cylinder
- REQ2 - The volume of the engine is 1600cc
- REQ3 - Each cylinder has four valves
- REQ4 - The injection system is governed by a the engine manager
- REQ5 - The engine provides 140ch at 6700 rpm/min
- REQ6 - The engine provides a torque of 150Nm at 5200 rpm/min
- …

Let’s build the requirement model corresponding to the specification

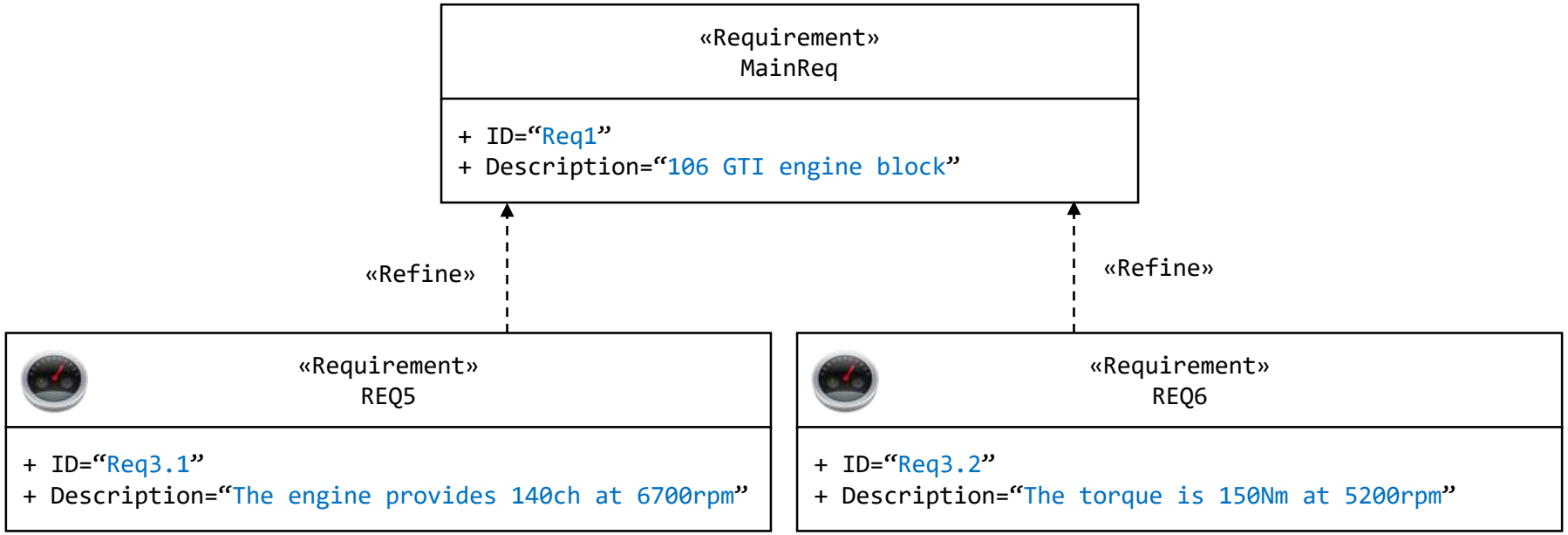
Capture *structural* requirements



Capture *functional* requirements



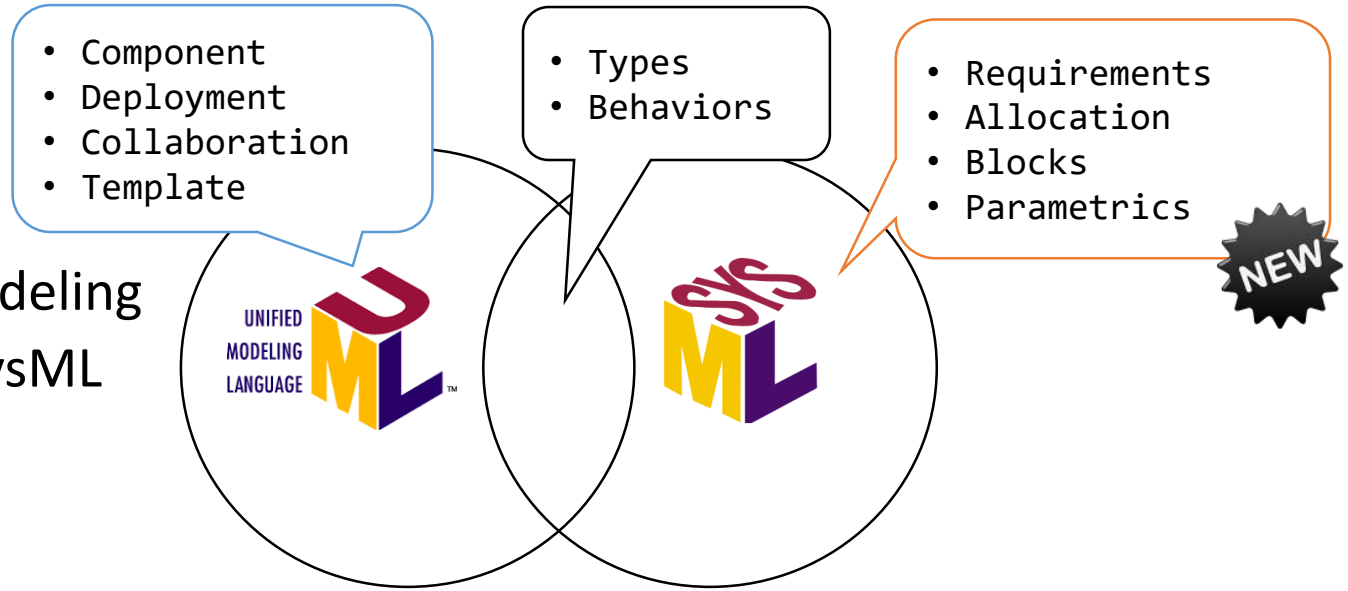
Capture *performance* requirements



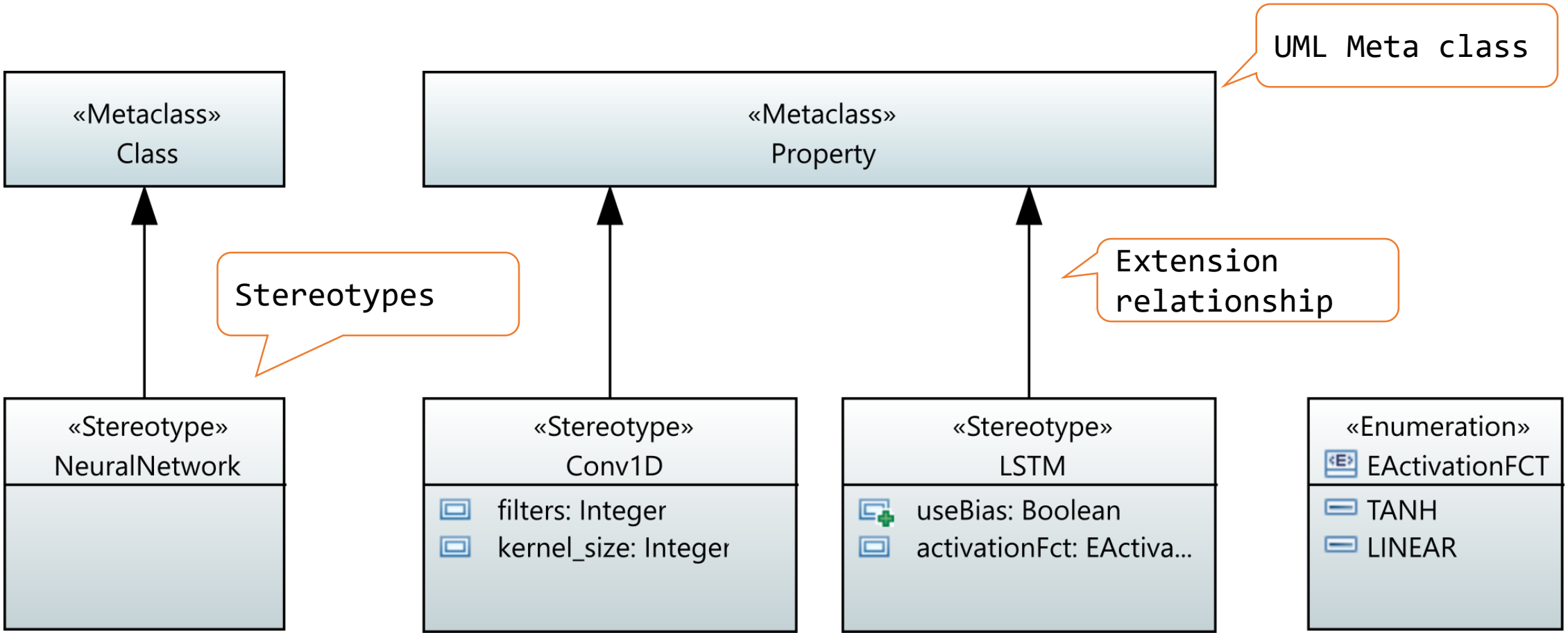
Interest requirement model / related work in SysML

- Interest of the requirements model
  - Capture and organize efficiently the requirements of a system
  - Link the requirements to the model of the system
    - What are the part of the system that satisfy the requirements
  - Link the requirements to the test cases validating the system
    - What are the test cases that ensure the requirements are verified

- SysML
  - Normative UML profile
  - Specialization of UML for system modeling
  - The addition of UML compared to SysML

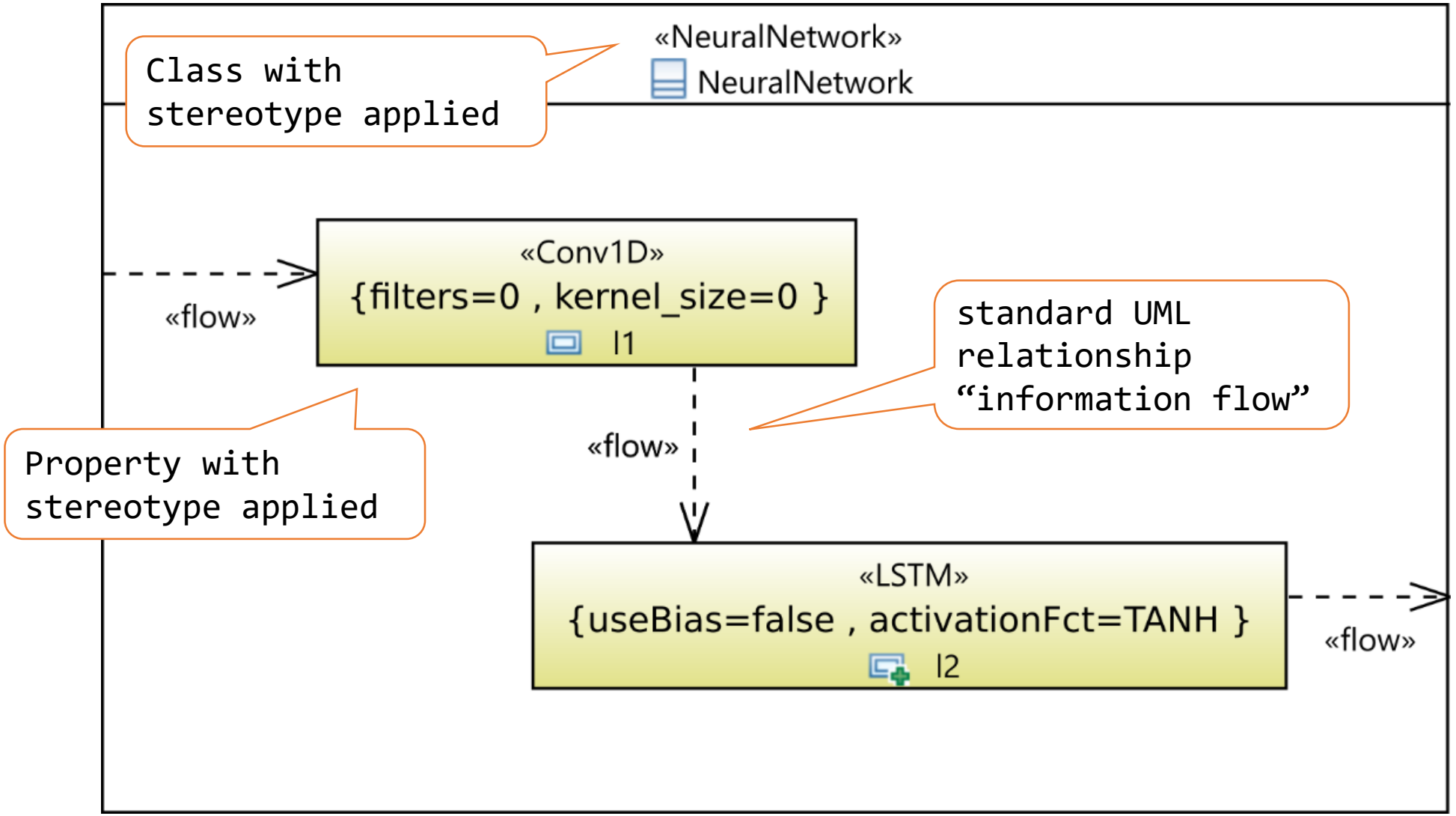


Simple Neural Network profile



Sample model applying the profile

Composite structure diagram
CSS (style sheet) modifies appearance



CSS – Cascading style sheet

```
Class {
   fillColor: white;
}
Property {
   maskLabel : name;
   fillColor: #e0e0a0;
}
Property > Compartment[kind="internalstructure"] {
   visible : false;
}
Compartment[type=StereotypeBrace] {
   visible : true;
}
```

Background color of compsite class is white

Show only name as label (not visibility,type, …)

Hide “internal structure” compartment

Stereotype attributes in Braces { … } after stereotype name

Exercise

- Create profile with Papyrus
- Create a new model, apply the profile and style-sheet
- Create the layers and flow